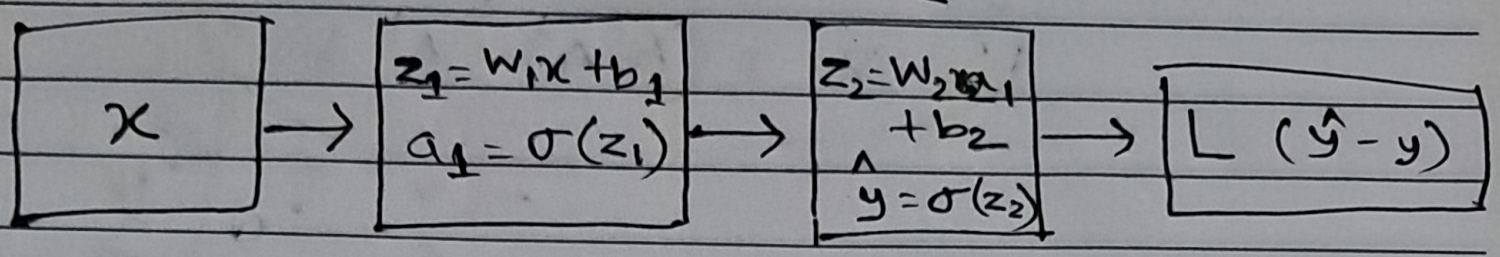
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For MSE Loss,

$$L = \frac{1}{m} \sum_{k=1}^{m} (\hat{y}_k - y_k)^2$$

where, m = output dimensions

$\hat{y}$ = predicted

y = actual

$$\frac{dL}{d\hat{y}_k} = \frac{2}{m}(\hat{y}_k - y_k)$$

For sigmoid activation,

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{d\sigma}{dz} = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1+e^{-z}-1}{1+e^{-z}} \times \frac{1}{1+e^{-z}}$$

$$= \{1 - \sigma(z)\}\sigma(z)$$

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Now, backpropagating from output layer

To see how the weight of layer 2 change and bias we need to follow the chain rule. As we cannot directly differentiate loss wrt. the weight. Rather, we'll continuously calculate the gradient of the variables that we have to ultimately satisfy our required rate of change.

$$\frac{\partial L}{\partial W_{2,kj}} = \frac{\partial L}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_{2,k}} \cdot \frac{\partial z_{2,k}}{\partial W_{2,kj}}$$

from layer k to j

$$= \frac{2}{m} (\hat{y}_k - y_k) \cdot \hat{y}_k (1 - \hat{y}_k) \cdot a_{1,j}$$

$$\frac{\partial L}{\partial b_{2,k}} = \frac{\partial L}{\partial \hat{y}_k} \cdot \frac{\partial \hat{y}_k}{\partial z_{2,k}} \cdot \frac{\partial z_{2,k}}{\partial b_{2,k}}$$

$$= \frac{2}{m} (\hat{y}_k - y_k) \cdot \hat{y}_k (1 - \hat{y}_k) \cdot 1$$

Taking $\frac{2}{m} (\hat{y}_k - y_k) \cdot \hat{y}_k (1 - \hat{y}_k) = \delta_{2,k}$

$$\frac{\partial L}{\partial W_{2,kj}} = \delta_{2,k} \, a_{1,j}$$

$$\frac{\partial L}{\partial b_{2,k}} = \delta_{2,k}$$

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Now backpropagate to find the gradients of layer j,

$$\frac{\partial L}{\partial W_{1,ji}} = \frac{\partial L}{\partial \hat{y}_u} \cdot \frac{\partial \hat{y}_u}{\partial z_{2,u}} \cdot \frac{\partial z_{2,u}}{\partial a_{1,j}} \cdot \frac{\partial a_{1,j}}{\partial z_{1,j}} \cdot \frac{\partial z_{1,j}}{\partial w_{i,ji}}$$

$$= \frac{2}{m}(\hat{y}_u - y_u)\ \hat{y}_u(1-\hat{y}_u) \cdot w_{2,uj} \cdot a_{1,j}(1-a_{1,j}) \cdot x_i$$

$$\frac{\partial L}{\partial b_{1,j}} \quad \frac{\partial L}{\partial b_{1,j}} \quad \frac{\partial L}{\partial \hat{y}_u} \cdot \frac{\partial \hat{y}_u}{\partial z_{2,u}} \cdot \frac{\partial z_{2,u}}{\partial a_{1,j}} \cdot \frac{\partial a_{1,j}}{\partial z_{1,j}} \cdot \frac{\partial z_{1,j}}{\partial b_{1,j}}$$

$$= \frac{2}{m}(\hat{y}_u - y_u)\ \hat{y}_u(1-\hat{y}_u) \cdot w_{2,uj} \cdot a_{1,j}(1-a_{1,j})$$

Taking this equal to δ_1

$$\frac{\partial L}{\partial w_{1,ji}} = \delta_{1,j} x_i$$

$$\frac{\partial L}{\partial b_{1,j}} = \delta_{1,j}$$

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Example

let, $x = 1$

$y = 1$

$w_1 = 0$

$w_2 = 1$

$b_2 = -1/2$

$b_1 = 0$

initially ;

$z_1 = 0$

$z_2 = 1 \times 1/2 - 1/2 = 0$

$a_1 = 1/2$

$\hat{y} = 1/2$

$L = (1 - 1/2)^2 = 1/4$

we calculate δ_1 and δ_2 ,

$$\delta_2 = \frac{2}{m} (\hat{y}_k - y_k) \hat{y}_k (1 - \hat{y}_k)$$

$$= 2 (1/2 - 1) \; 1/2 \; (1 - 1/2) = -1/4$$

$$\delta_1 = \frac{2}{m} (\hat{y}_k - y_k) \hat{y}_k (1 - \hat{y}_k) \cdot w_{2,kj} \cdot a_{ij} (1 - a_{ij})$$

$$= 2 \times (1/2 - 1) \cdot 1/2 \; (1 - 1/2) \cdot 1 \cdot 1/2 \; (1 - 1/2)$$

$$= 2 \times -1/2 \times 1/2 \times 1/2 \cdot 1/2 \times 1/2$$

$$= -1/16$$

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So, $\frac{\partial L}{\partial w_2} = \delta_2 \times a_1 = -\frac{1}{4} \times \frac{1}{2} = -\frac{1}{8}$

$\frac{\partial L}{\partial b_2} = \delta_2 = -\frac{1}{4}$

$\partial L / \partial w_1 = \delta_1 \times x = -\frac{1}{16}$

$\partial L / \partial b_1 = \delta_1 = -\frac{1}{16}$

This changed, accompanied by a suitable learning rate, have to be ~~multip~~ subtracted from previous weight and bias is the new set of weight and bias. This process needs to be carried out continuously and iteratively until any terminal condition is reached.